

Facial Recognition System Using AI and Python

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1. Introduction

The goal of this project is to build a facial recognition system that can identify individuals from images using artificial intelligence techniques. The system spans the full pipeline from raw data collection through preprocessing, feature extraction, and classification. A facial recognition system is a technology that can identify or verify a person by analyzing patterns in their face. It typically works by detecting a face in an image, extracting numerical features (called embeddings) that capture the unique geometry of the face, and then comparing those features against known individuals using a classification model. Facial recognition has broad real-world applications including security and surveillance, phone and device unlocking (such as Apple's Face ID), law enforcement identification, airport check-in automation, social media photo tagging, and access control for restricted areas. These systems rely on a combination of computer vision and machine learning to function accurately and efficiently.

2. Deliverable Summary

Deliverable 1 focused on data collection and preprocessing. The team collected images of 10 well-known celebrities from publicly available sources and organized them into a structured dataset with train, validation, and test splits. A preprocessing script was written to resize all images to 224x224 pixels, convert them to RGB color space, and normalize pixel values to a 0-1 range. The preprocessed images were saved as NumPy arrays for efficient downstream use.

Deliverable 2 focused on face detection, extraction, alignment, preprocessing, and embedding generation. Using dlib's HOG-based frontal face detector and 68-point facial landmark predictor, the pipeline detected faces in each image, cropped them with padding, aligned them by leveling the eyes using an affine rotation, resized them to a uniform 224x224, and generated 128-dimensional face embeddings using the face_recognition library. The outputs included aligned face images, a compressed embeddings file (face_embeddings.npz), a failure report, and five sample visualizations showing each stage of the pipeline.

Deliverable 3 focused on recognition models and evaluation. Four classifiers were trained on the 128-dimensional face embeddings: Support Vector Machine (SVM), K-Nearest Neighbors (KNN), Logistic Regression, and Cosine Similarity-based nearest neighbor matching. Each model was evaluated on the held-out test set using accuracy, precision, recall, and F1-score.

3. Methodology

The dataset consists of images of 10 celebrities: Angelina Jolie, Beyonce, Jennifer Lawrence, Megan Fox, Neil Patrick Harris, Robert Downey Jr., Taylor Swift, Tom Hanks, Tom Holland, and Will Smith. A total of 106 images were collected and organized into three splits: 69

for training, 17 for validation, and 20 for testing. Each split contains one folder per individual, with approximately 6 to 9 training images per person and 1 to 3 images in validation and test.

Embedding generation was handled by the `face_recognition` library, which uses a deep neural network (based on the ResNet architecture) to produce a 128-dimensional vector for each face. These embeddings capture the essential geometric and structural features of each face in a compact numerical representation. Before embedding generation, each image passed through the full preprocessing pipeline: face detection, bounding box extraction with padding, eye-based alignment using `dlib`'s 68-point landmark predictor, resizing to 224x224, and pixel normalization. Of the 106 images, 104 successfully produced embeddings. One image (Tom Hanks, training set) failed face detection entirely, and one image (Will Smith, validation set) failed at the embedding stage. The final embedding counts were 68 training, 16 validation, and 20 test samples.

The training embeddings and labels were used to fit each classifier, the validation set was used to confirm performance during development, and the test set served as the final unseen evaluation set.

Four models were implemented. The SVM used a linear kernel with regularization parameter $C=1.0$ and was trained on the 128-dimensional embeddings directly. KNN used $k=3$ neighbors with Euclidean distance as the similarity metric. Logistic Regression used a multinomial formulation with a maximum of 1000 iterations. The Cosine Similarity approach computed the cosine similarity between each test embedding and all training embeddings, then assigned the label of the most similar training sample (effectively a 1-nearest-neighbor classifier using cosine distance).

4. Results & Comparison

All four models achieved 100% accuracy on both the validation and test sets. The results are summarized below.

Model Performance Summary:

Model	Validation Accuracy	Test Accuracy
SVM (Linear Kernel)	100%	100%
KNN (k=3, Euclidean)	100%	100%
Logistic Regression	100%	100%
Cosine Similarity (1-NN)	100%	100%

Per-class precision, recall, and F1-score were all 1.00 for every individual across all four models. The confusion matrices for all models were perfect diagonal matrices with zero off-diagonal entries, meaning no misclassifications occurred for any celebrity in the test set.

The perfect accuracy across all models is a direct consequence of the high quality of the face embeddings combined with the small scale of the dataset. The `face_recognition` library's underlying deep network produces embeddings that are highly discriminative, meaning the 10 individuals in this dataset are well-separated in the 128-dimensional embedding space. With only 10 classes and 20 test samples, even simple classifiers like KNN and cosine similarity can distinguish between them without error.

Because all models performed identically, there is no single "best" model in terms of raw accuracy. However, in a practical sense, SVM and Logistic Regression are generally more

scalable to larger datasets and more robust to noise, while KNN and Cosine Similarity are simpler to implement and interpret. For this dataset size, all four are equally effective.

There were no common misclassifications to report. However, it is important to note that this level of performance is unlikely to hold on a larger, more challenging dataset with more individuals, greater variation in lighting and pose, or lower-quality images.

5. Challenges Faced

One challenge was inconsistent folder naming across the train, validation, and test splits. For example, some folders used abbreviated or slightly different names for the same person (e.g., "Beyonce" vs. "Beyonce T" vs. "Beyonce V," or "RDJ" vs. "Robert Downey Jr T"). This had to be resolved with a name normalization map so that labels matched correctly across all splits.

The dataset is quite small, with only 6 to 9 training images per person. While this was sufficient for the embedding-based approach to achieve perfect accuracy, it would not generalize well to a production system. Small datasets are prone to overfitting and do not capture the full variability of a person's appearance across different conditions.

Two images failed during the pipeline. Tom_Hanks_0001.jpg (training set) failed at the face detection stage, likely due to an unusual angle, occlusion, or low contrast that prevented dlib's HOG-based detector from finding a face. Will_Smith_0003.jpg (validation set) passed all stages up through alignment and preprocessing but failed at the embedding generation step, meaning the face_recognition library could not produce a valid encoding from the processed image.

File and path issues also came up during collaboration. The team worked across different environments (local machines, Google Colab, GitHub), and differences in file paths, zip structures, and operating systems occasionally caused scripts to fail until paths were standardized.

The 100% accuracy itself, while a positive result, also represents a limitation. The dataset is not large or diverse enough to truly stress-test the models, so the results should be interpreted as a proof-of-concept rather than evidence that these models would work in a real-world deployment.

6. Ethical Reflection

Facial recognition technology raises significant ethical concerns. Bias and fairness are major issues: many face recognition models have been shown to perform worse on people with darker skin tones, women, and other underrepresented groups. Our dataset of 10 celebrities is not demographically balanced, and any system trained on a small, non-representative dataset risks reinforcing existing biases. A more responsible approach would involve testing across a diverse range of ages, ethnicities, genders, and lighting conditions.

Privacy and consent are also critical. The images used in this project are of public figures sourced from publicly available images, which partially mitigates consent concerns for an educational project. However, deploying facial recognition on everyday people without their knowledge or explicit consent is a serious violation of privacy. Laws like the GDPR in Europe and BIPA in Illinois have begun to regulate how biometric data can be collected and used.

The potential for misuse is real. Facial recognition can be used for mass surveillance, tracking individuals without their knowledge, or enabling authoritarian control. Even well-intentioned deployments (like finding missing persons) can be repurposed for harmful ends if proper safeguards are not in place.

Finally, the reliability of AI systems matters. Our models achieved perfect accuracy on a small test set, but that does not mean they are infallible. False positives (misidentifying someone) and false negatives (failing to recognize someone) both have consequences, especially in law enforcement contexts where a wrong identification could lead to wrongful arrest. Any real-world deployment must account for error rates and include human oversight.

7. Team Contribution

Jarely Orozco managed the GitHub repository, organized the initial dataset structure with the train/validation/test splits, and coordinated deliverable submissions. Alejandra Garcia assisted with data collection, gathered celebrity images, and contributed to documentation and the README. Ahmad Barakat authored the Deliverable 1 preprocessing script (FacialRecognition.py) and the Deliverable 2 feature extraction pipeline (FeatureExtraction.py), including face detection, alignment, and embedding generation. Joe Castillo confirmed the dataset structure, ran the full Deliverable 2 pipeline end-to-end, collected the failure report, generated the five sample visualizations, and implemented the Deliverable 3 recognition models (SVM, KNN, Logistic Regression, Cosine Similarity).

8. Conclusion & Future Work

This project demonstrated a complete facial recognition pipeline from raw image collection to identity classification. The key takeaway is that high-quality face embeddings, generated by a pre-trained deep neural network, make the downstream classification task straightforward even for simple models. All four classifiers achieved perfect accuracy on the 10-person test set, validating the effectiveness of the embedding-based approach.

Several improvements could strengthen this system. First, expanding the dataset to include more individuals (50 or more) with greater variation in pose, lighting, expression, and image quality would provide a more realistic test of model performance. Second, incorporating deep learning classifiers such as a fine-tuned convolutional neural network or a siamese network for one-shot learning could improve robustness. Third, implementing real-time recognition using a webcam feed would move the project from a batch processing pipeline to an interactive application. Fourth, adding data augmentation (random crops, flips, brightness adjustments) during training could help the models generalize better from limited data. Finally, testing on a more demographically diverse dataset and reporting per-group accuracy would address fairness concerns and produce a more honest assessment of the system's real-world readiness.